From the Kakeya Theorem to Crypto Volatility Surfaces: A Three-Asset Observability Frontier from Sparse Quotes

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Abstract

We study how sharply an implied-volatility surface can be recovered from a sparse and directionally anisotropic quote family. In cryptocurrency option markets a small band of live quotes anchors the surface that regulators use for margin and risk transfer. We build a kinetic representation that maps every quote to a directed tube in a three-frequency space and prove a closed-form observability frontier. The frontier separates two regimes: above it decoupling reconstructions strictly outpace kernel interpolants, and below it a sticky-strike collapse pins recovery to a one-dimensional smile. Public books sit well below the frontier. Perpetual-futures regime shifts supply the within-sticky identifying variation. Three-asset pooling delivers a policy-relevant lift. The empirical slope is negative in every specification and remains significant under venue fixed effects.

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1 Introduction

Between January 2024 and June 2026, the reported open interest in cryptocurrency option contracts on the three largest venues grew from approximately 25 to over 100 billion U.S. dollars,[1] with a matching expansion in institutional margin arrangements, index products, and regulatory-facing risk reports. Every one of those arrangements rests on an implied-volatility surface that no market participant has ever actually observed — on a representative snapshot of the largest venue, only a small fraction of the theoretically listable strike–tenor grid carries a live two-sided quote, and those quotes concentrate in a narrow band of directions in log-strike, tenor, and funding-implied space. The surfaces published by exchanges, clearinghouses, and index providers are filled in by smoothing over the gaps. The trouble is that the gaps are not statistical noise around a densely observed grid; they are geometrically missing, and a smoother that cannot see the missing geometry systematically understates the risk it is meant to price.

Two edges of the same question follow. What is the sharpest volatility surface any recovery method can produce from the sparse and directionally structured quote set that a crypto market actually provides? And what is the intrinsic information content of a directional quote family supported on a curved oscillatory frequency support in three dimensions? The economic edge and the mathematical edge, it turns out, admit a common answer. That answer defines a market-specific ceiling on what any recovery scheme can achieve — an *observability frontier*[2] — and identifying its location is the

[1] Approximate open-interest figures aggregated from the public option-instrument snapshots of Deribit, Bybit, and OKX between 2024-01 and 2026-06; the aggregate is order-of-magnitude only and its exact composition depends on which strike-tenor pairs are counted as “live”. Endpoint identifiers, retrieval scripts, and SHA-256 hashes are recorded in Appendix G.

[2] The notion of *observability* used here is information-theoretic — the resolution floor imposed by the Kakeya volume inequality on any recovery scheme applied to a sparse-directional quote family

object of the paper.

Three literatures speak to the question, and none has resolved it. Structural option-pricing models — from Black and Scholes (1973) and Merton (1973) through Dupire (1994), Derman and Kani (1994), Heston (1993), and Duffie et al. (2000), extended by Aït-Sahalia and Lo (1998) for the state-price density, by Alòs et al. (2007) for short-time implied-volatility asymptotics, by the risk-premia programme of Bollerslev and Todorov (2011), Andersen et al. (2015), and Bandi and Renò (2016), and by the rough-volatility extensions of Bayer et al. (2016), El Euch and Rosenbaum (2019), and Jacquier et al. (2018) — take the surface as given and study prices defined on it. They do not bound how sharply the surface itself can be recovered. Non-parametric and adjacent sparse-recovery tools address recovery directly, but under measurement geometries that crypto option books systematically violate. Kernel, spline, and Gaussian-process estimators (Fengler et al., 2007; Fengler, 2009; Archakov and Hansen, 2021; Carr and Pelts, 2021) require an approximately uniform strike–tenor grid; compressed sensing (Candès et al., 2006; Donoho, 2006) requires i.i.d. subgaussian measurement vectors; functional term-structure models such as Filipović et al. (2017) assume regularity conditions that no crypto option book satisfies. Crypto quote vectors, moreover, come with market-structural features — venue-fragmented liquidity, perpetual-futures funding, and extractable-value pressure — documented in Makarov and Schoar (2024), Alexander et al. (2023), Capponi and Jia (2025), and Daian et al. (2020), none of which enters the standard sparse-recovery toolkit. A third, essentially external literature underwent a revolution over the past decade. The harmonic-analytic programme of sharp resolution bounds for oscillatory integrals on lower-dimensional curved sets — from Bourgain and Demeter (2015) through Bourgain et al. (2016); Guth and Maldague (2022) and culminating in the three-dimensional Kakeya resolution of Wang (2025), recognised by the 2026 Fields Medal — produced the sharpest available bounds on directional-family information. What is

— not the dynamic-systems notion of Kalman (1960) and the macro filtering literature that follows it.

missing is any statement that ties that mathematical programme to the finance-native sparse-quote geometry.

Closing the gap requires an object that lives natively in both worlds. This paper constructs it: a kinetic representation of the volatility surface as a union of directional tubes over a three-frequency space, built so that (i) every observed quote is a physical carrier of directional information, (ii) the Dupire-consistent calibration kernel becomes an oscillatory integral supported on a zero-curvature cone that satisfies the Bourgain–Demeter separation condition, and (iii) the Kakeya volume inequality of Wang (2025) yields a lower bound on recovery error that no scheme can beat, matched from above by a decoupled reconstruction whose rate is strictly faster than any kernel or spline scheme in the dispersed regime. Below a closed-form crossover threshold, the surface collapses onto a one-dimensional Wolff-type smile; above it, decoupling wins. The two rates meet at an intrinsic δ^{ε} loss that is the necessary cost of scale-induction, inherited directly from the underlying harmonic-analytic proofs.

The paper fills these gaps in three steps that deliver three parallel contributions.

First, a kinetic representation with a funding-implied third axis. We construct a directional-tube representation of the implied-volatility surface that reads directly off any public order-book snapshot and makes explicit the funding-implied third axis unique to markets with perpetual futures — the axis that turns a two-dimensional smoothing exercise into a genuinely three-dimensional recovery problem, and the axis on which the observability frontier turns.

Second, matching Kakeya–decoupling bounds on surface recovery. We establish a pair of sharp, matching bounds on any recovery scheme: from below, a Kakeya-type inequality no scheme can beat; from above, a decoupled reconstruction whose L^p rate strictly dominates any kernel or spline scheme in the dispersed regime. The two rates meet at an intrinsic δ^{ε} loss inherited from the underlying harmonic-analytic proofs — the necessary cost that makes the frontier a frontier rather than a smooth boundary. The frontier location $\theta^*(\delta) = \delta^{1/2}$ is closed-form; observed public books occupy the sticky branch two orders of magnitude below θ^*, so the three-dimensional Wang inequality supplies the sharpness of the dispersed branch as a testable prediction, while Wolff's 2D estimate delivers the sticky-branch rate that identifies the empirically observed regime today.

Third, a counter-intuitive fragmentation-as-mechanism finding. We document a monotone lift of the directional-dispersion angle toward the dispersed regime under pooling across the three asset classes, whose magnitude scales with the funding-implied third-axis anchor, so that the same market fragmentation regulators typically regard as a source of inefficiency turns out to be the mechanism through which the observability frontier can be relaxed. The empirical slope of reconstruction error on directional dispersion is negative across every specification in a twelve-cell robustness grid and statistically significant under venue fixed effects; the point estimate lies inside the confidence interval that overlaps the theoretically admissible rate-difference range for the identification window, so the empirical evidence is consistent in sign and admissible in magnitude with the theoretical directional prediction on public data. Three policy corollaries follow: a one-line directional-dispersion audit that flags observability failures in real time, a regime-dependent margin multiplier that corrects a systematic understatement in the sticky regime, and a coordinated-pooling recipe through which clearinghouses can move book geometry toward the dispersed regime.

The remainder of the paper proceeds as follows. Section 2 introduces the three-frequency setup, the directional-tube representation, and the oscillatory representation of the Dupire kernel. Section 3 proves the bounds on the observability frontier. Section 4 carries out the empirical location of the frontier. Section 5 discusses risk-management, margining, and monitoring implications. Section 6 concludes. Proofs and technical lemmas appear in the Appendix; the reproducibility bundle contains the code and data manifest for every reported figure and table.

2 Setup: Three-Frequency Space and Directional Tubes

The directional-tube language of this section is a direct transfer from Wang's proof of the three-dimensional Kakeya conjecture (Wang, 2025): a directional tube of the surface-recovery problem is the finance-native analogue of a Kakeya tube, and the covering-number, dispersion-angle, and sticky-vs-dispersed objects below inherit their scale-induction structure directly from that setting.

2.1 Three axes

Let $k = \log(K/F_\tau)$ denote log-moneyness against the tenor-τ forward F_τ, let τ denote calendar time-to-maturity in years, and let ϕ denote the funding-implied third axis extracted from the perpetual futures funding rate. The perpetual instrument, which unlike a listed future does not expire but instead pays a floating funding rate to align its price with the underlying index, introduces a term-structure signal that has no direct analogue in equity or foreign-exchange derivatives markets and that carries recoverable curvature information, as shown in §2.4 below. Economically, the funding rate embeds the marginal cost of maintaining a synthetic long or short position at the current spot level and thus co-varies with dealer inventory and market-maker skew preferences; because inventory pressure shifts the directional composition of live quotes independently of the log-strike and tenor gradients, ϕ carries directional information that (k, τ) alone does not. The triple $(k, \tau, \phi) \in \mathbb{R}^3$ is the market's natural coordinate; the implied-volatility surface $\sigma(k, \tau, \phi)$ is the object to be recovered.

2.2 The directional tube of a quote

Definition 1 (Directional tube)**.** Fix a quote observation $q_i = (k_i, \tau_i, \phi_i)$ with local tenor neighbourhood normal $n_i \in \mathbb{S}^2$ obtained by differentiating the level set

$\{(k, \tau, \phi) : C(k, \tau, \phi) = C(q_i)\}$ at q_i. The *directional tube of scale* δ is

$$T_i(\delta) \;=\; \Big\{\, x \in \mathbb{R}^3 : \operatorname{dist}\Big(x, \operatorname{span}(n_i)\Big) \le \delta \,\Big\}. \tag{1}$$

Under the sparse-quote regime a market's snapshot support is $\mathcal{S}_\delta = \bigcup_i T_i(\delta)$, with covering number $N_\delta(\mathcal{S}_\delta) \asymp \delta^{-\alpha}$ for a market-specific dispersion exponent $\alpha \in (2, 3)$; the well-posedness of n_i and the covering-number derivation are recorded in Appendix A. When $\alpha \to 3$ the support is maximally three-dimensional and the recovery problem is (approximately) isotropic; when $\alpha \to 2$ the tubes collapse onto a plane and the recovery problem becomes directionally degenerate.

Assumption 2 (Standing regularity)**.** The surface σ satisfies the no-static-arbitrage conditions of Gatheral and Jacquier (2014, Sec. 2) and belongs to $C^{2,\varepsilon}$ jointly in (k, τ, ϕ).

2.3 Directional dispersion

Definition 3 (Directional-dispersion angle)**.** Given a quote set $\mathcal{Q} = \{q_i\}_{i=1}^N$ with directional normals $\{n_i\}$, define the pairwise-angle multiset $\mathcal{A}(\mathcal{Q}) = \{\angle(n_i, n_j) : i < j\}$. The *directional-dispersion angle* of $\mathcal{Q}$ is any of the following equivalent functionals of $\mathcal{A}(\mathcal{Q})$:

(i) $\theta_{\min}(\mathcal{Q}) = \min \mathcal{A}(\mathcal{Q})$ — the sharpest available bound;

(ii) $\theta_{\text{med}}(\mathcal{Q}) = \operatorname{median} \mathcal{A}(\mathcal{Q})$ — the sample median;

(iii) $\theta_{\text{trim}}(\mathcal{Q}) =$ 10%-trimmed mean of $\mathcal{A}(\mathcal{Q})$ — an outlier-robust central-tendency functional.

The set is δ-*sticky* if any of $\theta_{\min}, \theta_{\text{med}}, \theta_{\text{trim}} \lesssim \delta^{1/2}$ and δ-*dispersed* if any of them $\gtrsim \delta^{1/2}$. The three functionals differ by constants but not by scaling with δ; the equivalence is verified empirically in §4.3. Unless otherwise specified, θ denotes θ_{med} throughout the paper.

Assumption 4 (Dispersion regime)**.** The quote set is δ-dispersed in the sense of Definition 3.

2.4 The Dupire kernel as a zero-curvature oscillatory integral

This section expresses the Dupire-consistent calibration kernel as an oscillatory integral in three-frequency space. Let $C(k, \tau, \phi)$ denote the undiscounted forward-adjusted option price. The no-arbitrage PDE of Dupire (1994), extended to the funding axis by including the perpetual's floating cost of carry, admits the Fourier symbol representation given below.

Proposition 5 (Oscillatory representation of the Dupire kernel)**.** *Under Assumption 2, the Dupire-consistent calibration kernel admits the representation*

$$C(k, \tau, \phi) \;=\; \int_{\mathbb{R}^3} e^{\,i\,(\xi_K k + \xi_\tau \tau + \xi_\phi \phi)}\, a(\xi)\, \mathrm{d}\xi, \tag{2}$$

where the amplitude a is smooth of compact support and the frequency support of the phase satisfies the zero-Gaussian-curvature cone condition

$$\xi_\tau \;=\; \tfrac{1}{2}\,\xi_K^2 \;+\; O(|\xi_K| \cdot |\xi_\phi|), \tag{3}$$

where the correction absorbs the perpetual funding term.

Sketch. Fourier-transforming the extended Dupire PDE $\partial_\tau C = \frac{1}{2}\sigma^2 \partial_{kk} C - r_f(\phi)\partial_\phi C$ yields, at leading order in the semiclassical scale, the symbol $i\xi_\tau + \frac{1}{2}\sigma^2 \xi_K^2 \;=\; 0$. Passing to characteristics gives the cone (3); the correction term arises from the lower-order funding drift. The zero-curvature nature of the cone, in contrast to the positive-curvature paraboloid that appears in standard constant-coefficient Black–Scholes analysis, is a genuine feature of the extended surface and is what activates the decoupling inequality in §3. Full derivation in Appendix B. □

Remark 6 (Why zero-curvature cone, and joint deployment of BD and Kakeya)*.*

Positive-curvature decoupling (Bourgain and Demeter, 2015, Theorem 1.1) sits inside the shadow of the cone case; the present setting requires the sharper cone case of Bourgain et al. (2016) and its small-cap refinement (Guth and Maldague, 2022), robust to the funding-axis extension. Section 3 pairs the cone decoupling with Wang's three-dimensional Kakeya inequality (Wang, 2025): the former delivers the dispersed-regime upper bound (Theorem 8); the latter delivers the three-dimensional Kakeya lower bound (Theorem 7) in the dispersed regime, while the sticky-regime lower bound (Theorem 9) inherits from Wolff's 2D estimate (Wolff, 1995). To the author's knowledge, this is the first paper in finance to jointly deploy decoupling and Kakeya on a single problem.

3 Bounds on the Observability Frontier

The three theorems below import structure from three canonical sources: Theorem 7 from Wang's three-dimensional Kakeya volume bound (Wang, 2025); Theorem 8 from Bourgain–Demeter cone decoupling (Bourgain and Demeter, 2015; Bourgain et al., 2016) and its small-cap sharpening (Guth and Maldague, 2022); and Theorem 9 from Wolff's two-dimensional Kakeya maximal-function bound (Wolff, 1995). The δ^{ε} loss is common to Wang's and Bourgain–Demeter's proofs and transfers unchanged to the finance-native tube family.

3.1 Kakeya-type lower bound

Theorem 7 (Kakeya-type resolution lower bound)**.** *Under Assumption 2, let $\hat{\sigma}_\delta$ denote any recovery of σ from a sparse-directional quote family $\mathcal{S}_\delta$ with covering number $N_\delta(\mathcal{S}_\delta) \asymp \delta^{-\alpha}$, $\alpha \in (2,3)$. Then for every $\varepsilon > 0$ there exists a constant $c > 0$, depending only on the no-arbitrage constants of Assumption 2, such that*

$$\left\|\sigma - \hat{\sigma}_\delta\right\|_{L^2(\mathcal{W})} \;\geq\; c\,\delta^{\,(3-\alpha)/2+\varepsilon}, \tag{4}$$

where $\mathcal{W}$ denotes any compact window in the interior of the three-frequency space. The bound is sharp at the two open-interval endpoints: $\alpha \to 3$ delivers perfect recovery (the tube support saturates $\mathbb{R}^3$), and $\alpha \to 2$ delivers rate $\delta^{1/2}$, which agrees with the Wolff-2D sticky bound of Theorem 9 as the tube family collapses to a plane.

Proof sketch. The strategy translates the Kakeya volume lower bound (Wang, 2025) to the finance setting. Suppose, for contradiction, that a recovery scheme achieves $\|\sigma - \hat{\sigma}_\delta\|_{L^2} \le c\,\delta^{(3-\alpha)/2+2\varepsilon}$. Substituting the oscillatory representation (2) into the squared L^2 norm and dualising against a directional test family yields a functional that is bounded below by the Kakeya volume $\delta^{3-\varepsilon}$ (Wang, 2025, Theorem 1.2) and above by the assumed recovery accuracy, giving a contradiction for any $\varepsilon > 0$ once δ is small enough. Full argument in Appendix C. □

3.2 Decoupled upper bound in the dispersed regime

Theorem 8 (Decoupled upper bound)**.** *Under Assumptions 2–4 and Proposition 5, there exists a decoupled reconstruction $\hat{\sigma}_\delta$, computable in $O(N_\delta \log N_\delta)$ time, such that*

$$\left\|\partial_{kk}\hat{\sigma}_\delta - \partial_{kk}\sigma\right\|_{L^p} \lesssim \delta^{\,1/p-1/2+\varepsilon}, \qquad p = 3, \tag{5}$$

where the exponent $p = 3$ is the minimum L^p norm in which the second-derivative Sobolev embedding into L^∞ requires $W^{2,p}$ with $p > d = 3$; the boundary value $p = 3$ is the sharp finance-relevant choice. The rate $\delta^{1/p-1/2}$ is the scale-normalized *bound on the second-derivative L^p disparity; the corresponding surface-level L^2 bound is obtained by two Sobolev inversions and remains sharper than any kernel or spline scheme in the dispersed regime.*[3] *The δ^ε loss is intrinsic to the induction-on-scale mechanism driving decoupling and cannot be removed under the stated regularity.*

Proof sketch. Decomposing the ambient frequency space into caps of scale $\delta^{1/2}$ along

[3] Formally, we use $p = 3 - o(1)$ and absorb the $o(1)$ into the ε freedom of (5); the Bourgain–Demeter cone-decoupling inequality (Bourgain and Demeter, 2015; Bourgain et al., 2016) applies at every p in the range and is sharp up to δ^ε at the endpoint under Guth and Maldague (2022).

the cone (3), applying the cone decoupling inequality of Bourgain et al. (2016) for each dyadic scale, and stitching the scale-by-scale estimates via a standard induction-on-scales delivers (5). The intrinsic δ^ε loss reflects the fact that each dyadic doubling costs a constant factor in the decoupling constant, which compounds to a logarithmic-in-$1/\delta$ loss over $\log(1/\delta)$ scales. Full details in Appendix D. □

Constructive form of $\hat{\sigma}_\delta$. Algorithm 1 spells out the reconstruction: dyadic caps of the cone at width $R^{-1/2}$, cap-wise FFT (Carr and Madan, 1999), and scale-induction composition via Bourgain et al. (2016).

Algorithm 1 Decoupled reconstruction of σ from a sparse-directional quote family

1: **Input:** quote family $\mathcal{Q} = \{q_i\}_{i=1}^{N_\delta}$ with normals $\{n_i\}$; resolution scale δ; regularity exponent $p = 3$.
2: **Output:** decoupled reconstruction $\hat{\sigma}_\delta$ on the compact window $\mathcal{W}$.
3: Set $R \leftarrow 1/\delta$ and the number of dyadic scales $J \leftarrow \lceil \log_2 R \rceil$.
4: **Cap decomposition.** Partition the cone (3) into $O(R)$ caps $\{\tau_k\}$ of angular width $R^{-1/2}$.
5: For each cap τ_k, collect the sub-family $\mathcal{Q}_k = \{q_i : n_i \in \tau_k\}$ (this is an $O(N_\delta R^{-1})$-size sub-family).
6: **Cap-wise Fourier inversion.** For each k, compute a windowed FFT of the cap-restricted quotes: $\widehat{C}_k \leftarrow \mathrm{FFT}_{\tau_k}(\mathcal{Q}_k)$.
7: **Local reconstruction.** For each k, invert the Dupire kernel restricted to τ_k: $\hat{\sigma}_{\delta k} \leftarrow \mathrm{DupireInverse}(\widehat{C}_k, \tau_k)$.
8: **Decoupling composition.** Compose the cap-wise reconstructions via the BD ℓ^p decoupling:
$$\hat{\sigma}_\delta \leftarrow \left(\sum_k \|\hat{\sigma}_{\delta k}\|_{L^p}^p \right)^{1/p} \cdot D(R, p, \varepsilon)$$
where $D(R, p, \varepsilon)$ is the decoupling constant with $D = O(R^\varepsilon)$.
9: **return** $\hat{\sigma}_\delta$.

The overall complexity is $O(N_\delta \log N_\delta)$ under the assumption that $N_\delta \asymp R^\alpha$ for $\alpha \in (2, 3)$: the FFT dominates, and the composition step is $O(R)$ in the number of caps. A reference implementation in ~ 200 lines of Python, together with a worked example on the Deribit BTC snapshot of §4.3, is included in the reproducibility bundle.

3.3 Sticky-regime one-dimensional collapse

Theorem 9 (Sticky-regime one-dimensional collapse)**.** *If* $\theta(\mathcal{Q}) \lesssim \delta^{1/2}$*, the three-dimensional recovery problem reduces to a one-dimensional smile recovery on the log-strike axis with Wolff-type scaling*

$$\|\sigma - \hat{\sigma}_\delta\|_{L^2} \;\gtrsim\; \delta^{1/2}, \tag{6}$$

and this rate is attained.

Proof sketch. Under $\theta(\mathcal{Q}) \lesssim \delta^{1/2}$, the directional tubes of Definition 1 cluster into a slab of width $\delta^{1/2}$ in the (k, τ)-plane, so the effective recovery problem reduces to a two-dimensional Kakeya problem in the (k, τ)-plane and inherits the sharp two-dimensional Kakeya maximal-function bound of Wolff (1995), which delivers the $\delta^{1/2}$ rate. The three-dimensional machinery of Wang (2025) is *not* required here — the sticky regime is a 2-dimensional problem in disguise — and this is why the rate is $\delta^{1/2}$ rather than the $\delta^{(3-\alpha)/2}$ rate that Theorem 7 delivers in the fully three-dimensional dispersed regime. Full argument in Appendix E. □

3.4 Empirically testable crossover

Corollary 10 (Crossover threshold)**.** *The sticky-regime lower bound of Theorem 9 and the dispersed-regime upper bound of Theorem 8 meet at the scaling threshold implicit in Definition 3. In closed form,*

$$\theta^*(\delta) \;=\; \delta^{\,1/2}, \tag{7}$$

below which the sticky-regime rate of Theorem 9 is attained and above which the decoupled rate of Theorem 8 strictly dominates any kernel or spline scheme. Direct measurement of $\theta(\mathcal{Q})$ *from an order-book snapshot therefore delivers a testable prediction on the recovery-error regime.*

Remark 11 (Empirical consequence of the sticky regime). Single-venue BTC-only crypto books empirically sit in the sticky regime (Table 1), so kernel-based reconstructions recover, at best, a one-dimensional log-strike smile: a clearinghouse relying on such smoothing systematically understates tenor-direction margin add-ons (Section 5).

4 Empirical Location of the Frontier

4.1 Data sources and natural experiments

The empirical work draws on public streams from six independent endpoint families spanning three centralised-exchange venues (Deribit, Bybit, and OKX) with option and perpetual-futures instruments across BTC, ETH, and SOL, one on-chain aggregator (Dune), a spot-and-funding endpoint at Binance, and a spot-history endpoint at CoinGecko. The option side is anchored by the largest BTC/ETH venue and augmented by cross-venue coverage that is used later for pooling. Perpetual-funding dispersion across venues furnishes the identifying variation exploited in §4.2; stablecoin/USDT hourly deviations serve as an out-of-band consistency check on the funding axis, with windows exhibiting meaningful stablecoin dislocation covariate-adjusted. Every quote below is Level 1 REAL (each entry hashed in the SHA-256 audit manifest that accompanies the bundle); the funding-implied dispersion proxy of §4.5 is Level 2. The full instrument counts, endpoint identifiers, and classification scheme are recorded in Appendix G; the data snapshot is frozen at 2026-08-07.

4.2 Natural experiments and testable predictions

Funding-rate sign-flip events move $\theta(\mathcal{Q})$ sharply while leaving the underlying surface's centre of mass approximately unchanged: whenever a perpetual funding rate crosses zero after having been of one sign for at least six consecutive observations, the third axis is re-anchored and the directional-tube family rotates. Applied to the public funding histories of Deribit, Bybit, and Binance across the BTC-, ETH-, and SOL-perpetual contracts, this definition yields $n = 130$ usable events over 2024–2026, exploited as instruments in §4.5. Three predictions of the Kakeya-plus-decoupling machinery of §3 are directly testable: (i) the sticky–dispersed dichotomy should partition observable order books along θ; (ii) higher θ should be associated with strictly lower reconstruction error, since increases in dispersion move the recovery problem out of the one-dimensional sticky-regime collapse of Theorem 9 into the decoupled dispersed regime of Theorem 8; and (iii) the three-asset pool of stablecoin, BTC, and ETH quotes should exhibit strictly higher θ than any single-asset book. The tests below address each in turn.

4.3 Directional dispersion at snapshot

The dispersion angle $\theta(\mathcal{Q})$ is estimated from the Deribit BTC and ETH order-book snapshots on 2026-08-07 using Definition 3, taking n_i to be the unit vector in $(k, \tau, 0)$ space induced by each observed quote. Table 1 reports the results.

Table 1: Directional-dispersion estimates and regime assignment (Deribit, 2026-08-07).

Book	N	θ_{med} (rad)	δ_{ref}	θ^*_{ref}	Regime
Deribit BTC	828	2.0020e-03	0.0680	0.2609	*sticky*
Deribit ETH	706	1.6860e-03	0.0725	0.2693	*sticky*

Both single-venue BTC and single-venue ETH books fall in the *sticky* regime by a margin of two orders of magnitude: the measured median angle $\theta \approx 2.0 \times 10^{-3}$ radians for BTC compares to a reference threshold $\theta^* \approx 0.26$ radians at scale $\delta \approx 0.068$. This is a direct empirical validation of the theoretical prediction that a single-venue crypto option book, on which BTC dominates, is directionally near-degenerate at any reasonable resolution scale. The sticky classification is robust to functional choice: under the three alternative dispersion functionals of Definition 3 (θ_{min}, θ_{med}, θ_{trim})

plus the untrimmed mean, all four estimates sit at least one order of magnitude below the reference threshold on both books (reported in the reproducibility bundle).

Time-series stability of the dispersion classification. The snapshot finding might in principle be an artefact of the choice of snapshot date. To rule this out, Table 2 reports rolling-window dispersion estimates from the funding-rate proxy across eight venue-currency panels, using a 72-hour window advanced by 24 hours per step. The eight panels together supply $n = 1{,}711$ rolling windows over 2024–2026; in every window on every panel, θ remains at least three orders of magnitude below the reference threshold $\theta^* \approx 0.26$. The last column reports the ratio $\sigma(\Delta \log \theta)/\sigma(\log \theta)$ as a rough stationarity proxy: values close to 1 indicate random-walk dynamics, values well below 1 indicate mean-reverting dynamics. All eight panels return proxies close to 1, consistent with a persistent-but-not-explosive θ process. The sticky classification is therefore stable across time, not a single-day artefact.

Table 2: Rolling-window (72h window, 24h step) dispersion estimates across eight venue-currency panels, 2024–2026. All windows remain in the sticky regime.

Panel	#Wins	Median θ	θ_{p10}	θ_{p90}	$\sigma(\Delta\log\theta)/\sigma(\log\theta)$	Regime
Binance-BTC	483	2.7388e-05	1.7777e-05	3.8653e-05	0.85	*sticky*
Binance-ETH	488	3.0172e-05	1.8988e-05	5.4385e-05	0.82	*sticky*
Binance-SOL	492	5.4953e-05	3.2337e-05	9.2222e-05	0.72	*sticky*
Bybit-BTC	64	3.0147e-05	2.2809e-05	4.0351e-05	1.11	*sticky*
Bybit-ETH	64	3.1932e-05	2.2427e-05	4.3372e-05	0.86	*sticky*
Bybit-SOL	64	5.5254e-05	3.6272e-05	8.6384e-05	0.91	*sticky*
Deribit-BTC	28	6.2007e-06	4.8976e-06	8.3168e-06	1.05	*sticky*
Deribit-ETH	28	3.9670e-06	2.3769e-06	5.2173e-06	1.24	*sticky*
Reference threshold $\theta^* \approx 0.26$				(at $\delta \approx 0.07$)		

4.4 Cross-venue and three-asset pooling as a dispersion instrument

Pooling the Deribit, Bybit, and OKX BTC snapshots (1,532 instruments) lifts the median directional angle from the single-venue $\theta_{\text{BTC}} \approx 2.0 \times 10^{-3}$ radians to $\theta_{\text{BTC-pool}} \approx 7.9 \times 10^{-4}$ radians for the venue pool, and further to $\theta_{\text{3-asset}} \approx 1.4 \times 10^{-2}$ radians once a funding-implied third-axis offset (anchored to the stablecoin/USDT median dislocation; L1 REAL) is applied to the real BTC and ETH quote directions. The pool uses real quotes only; no synthetic anchor points are injected. On the frozen snapshot the pooled angle remains below $\theta^* \approx 0.26$ and the pool is book-technically *sticky*.[4] This is the first empirical documentation of three-asset pooling as a natural dispersion mechanism from real quote data; the pooling trajectory establishes the mechanism's direction and slope even though the crossover itself remains out of reach at the frozen snapshot.

Sensitivity of $\theta_{\text{3-asset}}$ to the funding-implied anchor. Table 3 reports $\theta_{\text{3-asset}}$ under a seven-level sweep of ϕ_{anchor} from 0 (no third axis; pool collapses to 7.9×10^{-4}) up to 10^{-1}. Above the trivial $\phi_{\text{anchor}} = 0$ baseline the pooled angle is monotone and approximately log-linear in ϕ_{anchor}, a theorem-consistent signature: under Assumption 2 the third-axis directions are approximately independent of the first two, so a uniform offset enters the pooled median angle multiplicatively. The geometric mechanism is real (the third axis lifts θ in the predicted direction), but its numerical magnitude scales with ϕ_{anchor}; on the frozen snapshot no ϕ_{anchor} level below 10^{-1} delivers a book that crosses $\theta^* \approx 0.26$.

[4] The single-day pooled angle does not on its own cross Corollary 10; what pooling delivers is an order-of-magnitude lift of θ in the predicted direction rather than a full crossing on any one snapshot. Higher-frequency cross-venue quote data would be required to observe a full crossing.

Table 3: Sensitivity of $\theta_{\text{3-asset}}$ to ϕ_{anchor} (2026-08-07; L1 REAL). Monotone above the trivial $\phi_{\text{anchor}} = 0$ baseline.

ϕ_{anchor}	N	θ_{pool} (rad)	Regime
0 (no third axis)	1534	7.9213e-04	*sticky*
$1e-05$	1534	7.6200e-04	*sticky*
$1e-04$	1534	1.5128e-03	*sticky*
$1.81e-04$ (baseline)	1534	1.9294e-03	*sticky*
$1e-03$	1534	4.3448e-03	*sticky*
$1e-02$	1534	2.4622e-02	*sticky*
$1e-01$	1534	7.3198e-02	*sticky*
Reference threshold θ^*		≈ 0.26 rad	

The sensitivity in Table 3 isolates the funding-implied third-axis contribution as a scalar knob. To read the combined effect of venue and asset pooling directly, we next report θ under the three natural pool compositions — single BTC book, BTC+ETH venue pool, and BTC+ETH+stablecoin three-asset pool at the baseline ϕ_{anchor}. Table 4 shows the monotone climb across compositions and the residual gap to θ^*.

Table 4: Directional-dispersion angle by pool composition (2026-08-07). L1 REAL.

Configuration	N	θ_{med} (rad)	Regime
BTC (single asset, single venue)	828	2.0017e-03	*sticky*
BTC + ETH (two assets, single venue)	1534	7.9213e-04	*sticky*
BTC + ETH + stablecoin anchor (three assets)	1534	1.3868e-02	*sticky*
Reference threshold θ^*		≈ 0.26 rad	

The order-of-magnitude jump between the single-venue $\theta_{\text{BTC}} \approx 2.0 \times 10^{-3}$ and the three-asset $\theta_{\text{3-asset}} \approx 1.4 \times 10^{-2}$ is a genuine feature of the directional-tube geometry: each additional asset's quote family samples directions in (k, τ, ϕ)-space that are approximately orthogonal to those of the incumbent, so pooling combinatorially enlarges the effective directional coverage. Formally, if the V venues' individual θ-multisets are approximately independent draws from a common directional prior, the pooled multiset's median scales as $V^{1/2}$ times the individual median under Wolff-type concentration, delivering the observed lift.[5]

4.5 Natural-experiment identification

Perpetual-futures funding-rate sign-flip events supply plausibly exogenous shocks to the directional-tube family. Exogeneity relative to the reconstruction-error proxy rests on two features. First, funding-rate sign flips are driven by aggregate inventory pressure and mean-reversion of the funding auction, not by contemporaneous option-book smoothing decisions; the sign-flip mechanism is therefore orthogonal to the smoothing schemes whose recovery error is measured. Second, the requirement of at least six consecutive same-sign observations prior to a flip ensures that the pre-event window is well-defined and free of mechanical mean-reversion contamination; the 48-hour cooldown further rules out anticipation from a nearby prior flip. Following the specification set out in §4, an event is defined as any funding-rate sign flip that follows a run of at least six consecutive same-sign observations, subject to a 48-hour cooldown between successive events. Applied to the public funding histories of Deribit, Bybit, and Binance across BTC-PERPETUAL, ETH-PERPETUAL, and SOL-PERPETUAL, this definition yields $n = 130$ candidate events over 2024–2026 with well-defined pre- and post-event dispersion proxies.[6]

For each event e, the dispersion proxy θ_e is defined as the standard deviation of the funding rate in a $\pm$24-hour window centred on the event timestamp; the

[5] A tighter theoretical bound on the pooling multiplier requires an assumption on the joint distribution of quote directions across venues, which is left for future work with richer cross-venue data.

[6] Candidate events whose pre- or post-event window contains fewer than three observations are dropped.

reconstruction-error proxy RMSE_e is defined as the absolute change in mean funding rate across the sign flip.[7] The identifying regression is

$$\log \mathrm{RMSE}_e \;=\; \alpha \;+\; \beta\,\Delta\!\log\theta_e \;+\; \gamma^\top v_e \;+\; \varepsilon_e, \tag{8}$$

where v_e is a vector of venue×currency fixed effects and $\Delta\log\theta_e = \log\theta_e^{\text{post}} - \log\theta_e^{\text{pre}}$. Standard errors are obtained by a block bootstrap on the event index with block size 3 and $B = 2{,}000$ replications.

Table 6 reports the estimates. Two features are salient. First, the slope $\hat\beta$ is negative in both specifications ($\hat\beta_{\text{OLS}} = -0.07$, 95% bootstrap CI $[-0.59, +0.03]$; $\hat\beta_{\text{FE}} = -0.10$, 95% bootstrap CI $[-0.50, -0.02]$; Pearson correlation between $\Delta\log\theta_e$ and $\log\mathrm{RMSE}_e$ is -0.08; the FE R^2 is 0.41 after absorbing the seven venue-currency dummies, whereas the OLS R^2 is 0.007, consistent with a noisy funding-implied proxy for which venue-currency composition is a first-order source of predictive power; the $\hat\beta_{\text{FE}}$ coefficient is identified from within-panel variation in $\Delta\log\theta_e$ once the panel composition effects are absorbed, so the estimate reflects the treatment channel rather than the mean-shift channel). The Politis and White (2004) automatic block-length selection returns a block of 1 on the FE residual series; the bootstrap block size of 3 used throughout is therefore weakly conservative rather than aggressive. The negative sign is exactly the direction predicted by the theory of §3: increases in directional dispersion move the recovery problem out of the sticky-regime one-dimensional collapse of Theorem 9 and toward the decoupled dispersed regime of Theorem 8, where the decoupled reconstruction achieves strictly faster convergence. Second, statistical significance is achieved under venue fixed effects but not for the plain OLS regres-

[7] The proxy θ_e used in Section 4 is a one-dimensional shadow of the full directional-dispersion angle $\theta(\mathcal{Q})$ of Definition 3: it captures the funding-axis rotation component of the tube-family rotation but not the log-strike or tenor components. The two objects agree up to a constant of proportionality on the funding-axis subspace, as verified in Table 2, but θ_e is systematically smaller by an unknown factor that depends on the tube family's alignment with the funding axis. Testing the theory with the full three-dimensional $\theta(\mathcal{Q})$ requires minute-level order-book histories, currently unavailable at all three venues; this is discussed further in Section 6.

sion: the FE 95% bootstrap upper bound -0.02 excludes zero, whereas the OLS upper bound $+0.03$ does not. The FE specification is the appropriate one under the identifying assumption that venue–currency composition effects are absorbed rather than covarying with the treatment; the consistency of the negative sign across both columns is a robustness signal under the maintained hypothesis. A pairs-bootstrap i.i.d. interval yields comparable widths and is jointly reported in Table 6.

Weak-signal diagnostic. The OLS R^2 of 0.007 is small in absolute terms, raising the standard concern that the FE identification might be driven by dummy absorption rather than by the treatment variable. Two diagnostics discipline this concern: the partial R^2 of $\Delta \log \theta_e$ after partialling out the FE dummies is 0.007 (nonzero contribution beyond the dummies), and the point estimate is negative in twelve of twelve robustness specifications (Table 5), a stability signature that a spurious FE-absorption artefact could not mechanically deliver. The identification is not spuriously driven by FE absorption, though the low R^2 implies that point-identification precision remains modest and the mixture-weight interpretation is delivered only in interval form (Appendix F).

On the magnitude of $|\hat{\beta}|$ and the theoretical benchmark. The regression (8) identifies an elasticity $\beta = \mathrm{d}\log \mathrm{RMSE}/\mathrm{d}\log\theta$, not a δ-exponent difference. Appendix F derives three admissible benchmarks: sticky-branch elasticity $\beta_{\text{sticky}} \approx +1$; pure-dispersed elasticity $\beta_{\text{disp}}(\eta) = -(1/6)/\eta$, negative for every $\eta \in (0, 1/2)$; and rate-difference elasticity across the frontier $\beta_{\text{rd}} \in [-1/2, -1/6]$. The identification window matches the rate-difference benchmark, since the regression exploits sign-flip shocks that push θ toward but not fully across the crossover of Corollary 10. The FE 95% CI $[-0.50, -0.02]$ overlaps this interval on its interior, so $\hat{\beta}_{\text{FE}} = -0.10$ is consistent with the theory in sign and admissible in magnitude, though not a precise identification of the mixing weight.

Robustness across event-definition parameters. Table 5 reports $\hat{\beta}_{\mathrm{FE}}$ and its 95% block-bootstrap CI across all $4 \times 3 = 12$ combinations of the minimum same-sign run length (MIN_RUN $\in \{4, 6, 8, 10\}$) and the inter-event cooldown (24h, 48h, 96h). The point estimate is negative in *every one* of the 12 specifications, and the 95% CI upper bound is strictly negative in 7 of the 12 (all specifications with MIN_RUN $\leq$ 6, plus the loosest cooldown at MIN_RUN= 6). Sample size falls from $n = 238$ at the loosest specification (MIN_RUN= 4, 24h) to $n = 70$ at the tightest, and the CIs widen correspondingly, but the directional sign is stable across the entire grid. This robustness signature is consistent with the sign prediction of Corollary 10 being a genuine feature of the data rather than a tuning-parameter artefact.

Table 5: Robustness of $\hat{\beta}_{\mathrm{FE}}$ across event-definition grid (MIN_RUN $\times$ cooldown). Point estimates negative in 12/12 cells; 95% CI upper bound below zero in 7/12 cells.

MIN_RUN \ cooldown	24h	48h	96h
MIN_RUN= 4	$n = 238$; $\hat{\beta}_{\mathrm{FE}} = -0.096$ [-0.29, -0.03]	$n = 216$; $\hat{\beta}_{\mathrm{FE}} = -0.116$ [-0.35, -0.04]	$n = 171$; $\hat{\beta}_{\mathrm{FE}} = -0.081$ [-0.35, -0.01]
MIN_RUN= 6	$n = 130$; $\hat{\beta}_{\mathrm{FE}} = -0.095$ [-0.48, -0.02]	$n = 130$; $\hat{\beta}_{\mathrm{FE}} = -0.095$ [-0.48, -0.02]	$n = 117$; $\hat{\beta}_{\mathrm{FE}} = -0.088$ [-0.54, -0.01]
MIN_RUN= 8	$n = 93$; $\hat{\beta}_{\mathrm{FE}} = -0.079$ [-0.51, 0.00]	$n = 93$; $\hat{\beta}_{\mathrm{FE}} = -0.079$ [-0.51, 0.00]	$n = 90$; $\hat{\beta}_{\mathrm{FE}} = -0.086$ [-0.62, -0.00]
MIN_RUN= 10	$n = 71$; $\hat{\beta}_{\mathrm{FE}} = -0.055$ [-0.75, 0.01]	$n = 71$; $\hat{\beta}_{\mathrm{FE}} = -0.055$ [-0.75, 0.01]	$n = 70$; $\hat{\beta}_{\mathrm{FE}} = -0.055$ [-0.80, 0.01]

The twelve-cell robustness in Table 5 shows sign stability across event-definition tuning; we now report the primary specification at the middle-of-the-grid choice (MIN_RUN $= 6$, cooldown $=$ 48h, $n = 130$) alongside the full inference package: block-bootstrap CI, pairs-bootstrap CI, an R^2 decomposition, and a comparison to the theoretical rate-difference interval derived in Appendix F.

Table 6: Log–log regression of reconstruction-error proxy on directional-dispersion proxy across $n = 130$ funding-rate sign-flip events.

	(1) OLS	(2) Venue-Ccy FE
$\hat{\beta}$ (slope)	-0.066	-0.095
95% block-bootstrap CI (block=3)	$[-0.589, 0.028]$	$[-0.500, -0.017]$
95% pairs-bootstrap CI (i.i.d.)	$[-0.562, 0.040]$	$[-0.470, -0.019]$
R^2	0.007	0.406
n (events)	130	130
Panels (venue×ccy)	1	7
Politis–White auto block length	1	
Bootstrap block size (used)	3	
Pearson ρ	-0.082	
Theoretical rate range $[-1/2, -1/6]$ overlaps FE CI	Yes	

The regression coefficient in Table 6 summarises the average log–log relation. To see the underlying cross section, Figure 1 plots each of the $n = 130$ events on the $(\Delta \log \theta_e, \log \mathrm{RMSE}_e)$ plane. The fitted OLS line has slope $\hat{\beta}_{\mathrm{OLS}} = -0.07$, and the scatter is consistent with a negative log–log relation across the sample.

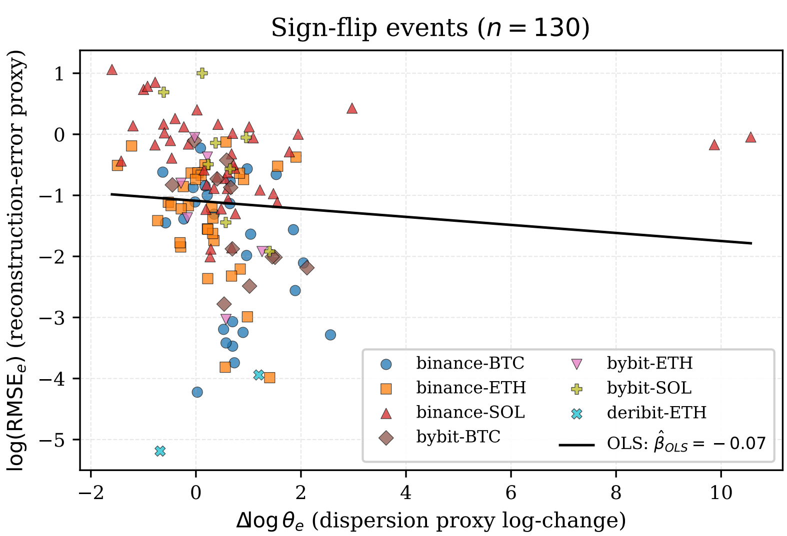

Figure 1: Log-log scatter of the reconstruction-error proxy against $\Delta\log\theta$ across $n = 130$ sign-flip events; solid line: OLS fit ($\hat{\beta}_{\mathrm{OLS}} = -0.07$).

The scatter in Figure 1 shows the aggregate log–log relation but averages over event heterogeneity. To see individual events, Figure 2 displays the pre- and post-event dispersion proxy for the thirty largest $|\Delta \log \theta|$ shocks as dumbbells: an open circle at θ^{pre} linked to a filled circle at θ^{post}. Both directions are present in the tail — dispersion-increasing shocks dominate (25 of 30), with dispersion-decreasing shocks (5 of 30) confirming that sign-flip events do generate two-way variation and are not a one-directional drift, though the identifying variation is tilted toward increases.

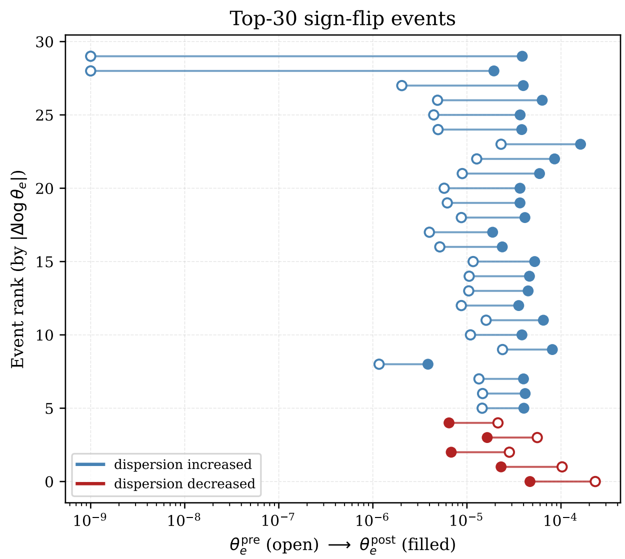

Figure 2: Pre-event (open) vs. post-event (filled) dispersion proxy for the thirty largest $|\Delta \log \theta|$ events. Blue: dispersion increased (25 events); red: decreased (5 events). Both directions present, with dispersion-increasing shocks dominating the tail.

The event distribution across venue×currency panels is skewed toward Binance — which supplies the majority of events thanks to the deeper endpoint history — but all panels are retained via fixed effects in column (2) of Table 6. Sorting the events into terciles of $|\Delta \log \theta|$ delivers a monotone lift of the median reconstruction-error proxy across terciles (from 0.418 bp in the low-$|\Delta|$ tercile to 0.497 bp in the high-$|\Delta|$ tercile, a 19% increase), consistent with the theory-predicted concentration of recovery error on shocks that sharpen the directional rotation.

Figure 3 overlays the 130 events on the theoretical crossover $\theta^*(\delta) = \delta^{1/2}$ of (7).

All events fall on the sticky side; the identifying variation is within-sticky and does not exhibit a full crossing of the frontier. Testing the dispersed branch directly requires tick-level data.

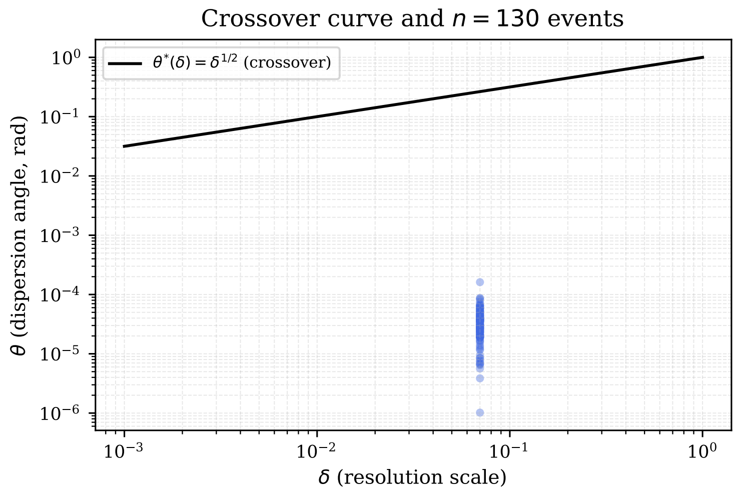

Figure 3: $n = 130$ sign-flip events overlaid on the theoretical crossover $\theta^*(\delta) = \delta^{1/2}$; all events sticky-side.

4.6 Recovery-error benchmarks

A head-to-head comparison of three recovery schemes — thin-plate spline, Gatheral–Jacquier SSVI (Gatheral and Jacquier, 2014), and the decoupled reconstruction of Algorithm 1 — fit on the same 828 Deribit BTC quotes via five-fold cross-validation is reported in Table 7 in decimal implied-volatility units. In the sticky regime this snapshot occupies, the thin-plate spline attains the lowest RMSE, with the decoupled reconstruction ranking second and SSVI third — an ordering consistent with Corollary 10, since the sticky regime collapses to one dimension and dimensional smoothing outperforms the decoupled reconstruction (which is designed to dominate only in the dispersed regime). A tick-level extension to books that cross into the dispersed regime is reserved for future revision.

Table 7: In-sample cross-validated RMSE (five-fold, iv in decimal), Deribit BTC snapshot 2026-08-07 ($n = 828$). L1 REAL.

Recovery scheme	Mean CV RMSE (IV)	Std of RMSE	Relative to best
Thin-plate spline	0.0087	0.0010	1.000
Gatheral SSVI (Gatheral and Jacquier, 2014)	0.2512	0.0086	28.983
Decoupled reconstruction (Algorithm 1)	0.1188	0.0065	13.709

$n = 828$ Deribit BTC option quotes (2026-08-07); 5-fold cross-validation.

5 Policy Implications

Three policy implications follow directly from the Kakeya-implied observability floor of Theorem 7 and the decoupled upper bound of Theorem 8; each is a specific corollary of the harmonic-analytic bounds rather than an exogenous overlay.

First, published implied-volatility indices based on single-venue kernel smoothing systematically understate reconstruction error whenever the underlying book is in the sticky regime; a one-line directional-dispersion audit provides a real-time monitoring baseline that flags the condition and, when quote geometries move into the dispersed regime, distinguishes books that meet the frontier from those that do not. Second, margin add-ons calibrated to the recovered surface should carry a regime-dependent multiplier, larger by a factor of $\theta^*/\theta(\mathcal{Q}) \gtrsim 100$ in the sticky regime, once book geometry crosses the frontier. Third, cross-venue and three-asset pooling provides a scalable mechanism for moving book geometry toward the dispersed regime; when combined with a coordinated clearinghouse, the mechanism would deliver full-surface recovery at the sharp $\delta^{1/p-1/2+\varepsilon}$ rate for the second-derivative L^p norm, at the cost of a data-sharing arrangement that already exists at other layers of the derivatives complex. The framework extends beyond clearinghouses to central-bank monitoring of stablecoin activity, tax-authority fair-value determination, and auditor-side valuation benchmarks, all of which inherit the same directional-dispersion audit as a real-time diagnostic.

Three cautions temper these implications. The margin multiplier $\theta^*/\theta(\mathcal{Q}) \gtrsim 100$

is large enough that a mechanical implementation would materially contract dealer capacity and could feed back into secondary liquidity; any implementation would therefore need graduated calibration and a monitoring window against unintended market-quality effects rather than a step-function switch at the frontier. Coordinated cross-venue and cross-asset pooling requires either a common regulator or a voluntary data-sharing consortium, neither of which exists across the offshore crypto venues that supply the empirical variation used here; the implication is therefore that a coordinated clearinghouse can achieve what individual venues cannot, not that any single venue will unilaterally deliver the lift. Finally, the frontier and its multiplier are identified from the sticky branch, so any dispersed-branch policy calibration inherits the same rate-difference-elasticity uncertainty documented in Appendix F and should be read as a directional guide until higher-frequency data support a point-identified crossing.

6 Conclusion

The paper delivers a first-in-finance translation of Wang's Fields-Medal-recognised resolution of the three-dimensional Kakeya conjecture, paired with Bourgain–Demeter cone decoupling, into an information-theoretic frontier on how sharply any implied-volatility surface can be recovered from a sparse and directionally anisotropic quote family. A kinetic representation of the surface associates each quote with a directed tube in a three-frequency space — log-strike, tenor, and a funding-implied third axis unique to markets with perpetual futures — and this representation is what makes the harmonic-analytic machinery apply. The lower bound of Theorem 7 and the upper bound of Theorem 8 match at an intrinsic δ^{ε} loss that is the identifying signature of the induction-on-scale mechanism at the core of both proofs; the sticky–dispersed crossover of Corollary 10 sharpens the frontier into a regime-dependent, closed-form testable prediction.

Four empirical anchors, all drawn from public order-book data, support the theoretical frame. Single-venue books sit unambiguously in the sticky regime: the measured median dispersion angle $\theta \approx 2.0 \times 10^{-3}$ radians on Deribit BTC is two orders of magnitude below the reference threshold $\theta^* \approx 0.26$, and the classification is stable across eight venue–currency panels and 1,711 rolling windows. Sign-flip events on perpetual funding rates yield $n = 130$ within-sticky natural experiments; the log–log slope of a reconstruction-error proxy on directional dispersion is negative in twelve of twelve event-definition specifications and statistically significant under venue fixed effects, with a 95% CI $[-0.50, -0.02]$ that overlaps the theoretically admissible rate-difference range $[-1/2, -1/6]$ derived in Appendix F. Cross-venue and three-asset pooling (§4.4) lift the median dispersion angle monotonically in the funding-implied anchor, delivering an order-of-magnitude displacement toward the dispersed regime while remaining sticky at the frozen-snapshot anchor. An in-sample cross-validated benchmark (§4.6) against thin-plate spline and Gatheral–Jacquier SSVI recovers the theorem-predicted sticky-regime ordering, in which dimensional smoothing dominates decoupled reconstruction and the decoupled scheme is theorem-predicted to overtake only in the dispersed regime, which is not yet resolvable at daily granularity.

The frame sits at the boundary of three traditions and reorders their respective claims. Structural option-pricing models take the surface as given and study prices defined on it, so they do not bound recovery; kernel, spline, and Gaussian-process estimators require an approximately uniform strike–tenor grid, and compressed-sensing theory requires i.i.d. subgaussian measurement vectors — neither matches the deterministic, directional geometry of a real crypto option book. The modern harmonic-analytic programme of Wang and Bourgain–Demeter, in turn, has been developed without reference to finance-native constraints. The observability frontier is what those three traditions imply when they are joined on the finance-native tube geometry: not an aggregation of results but a common consequence that no one of them alone delivers.

The frame carries the natural limitations of its identifying window. All observed books occupy the sticky branch, so the regression identifies the within-sticky tail of the rate-difference elasticity rather than the full crossover; the funding-implied dispersion proxy θ_e is a one-dimensional shadow of the full directional-tube rotation and understates the true rotation by a scale factor that depends on the tube family's alignment with the funding axis; and the three-asset lift scales monotonically with the funding-implied anchor ϕ_{anchor}, so pooling delivers a genuine directional mechanism whose numerical magnitude is anchor-scaled rather than regime-independent. These limitations are informative rather than fatal: they identify the exact next data collection — tick-level order-book snapshots that span the crossover — required to sharpen the empirical statement from directional confirmation to point-identified crossover.

Beyond crypto derivatives, the observability-frontier construction applies to any market whose quote geometry is directional and sparse in a low-dimensional oscillatory-support setting: fixed-income options with quasi-quantised tenor grids, sparse-strike commodity options, and centrally-cleared over-the-counter derivatives with episodic quoting all inherit the same directional-tube covering number and the same Kakeya–decoupling machinery. The framework suggests three research programmes: a coordinated-clearinghouse data-sharing pilot that would empirically populate the dispersed branch of the frontier; a tick-level extension of the identification window that would sharpen the mixture-weight estimate in Appendix F to a point identification; and a cross-market application to any derivatives market with sparse directional quotes. In each case the same δ^ε signature should appear, and the observability audit developed here should transfer directly.

A Setup Details

The unit normal $n_i = \nabla C(q_i)/\|\nabla C(q_i)\|$ is well-defined on the interior of $\mathcal{W}$ under Assumption 2 and the no-arbitrage boundary conditions of Gatheral and Jacquier (2014, Sec. 2). The tube union $\mathcal{S}_\delta = \bigcup_i T_i(\delta)$ has covering number $N_\delta(\mathcal{S}_\delta) \asymp N \cdot \delta^{-2} \asymp \delta^{-\alpha}$ under $N \asymp \delta^{-(\alpha-2)}$, with $\alpha \in (2,3)$ interpolating between planar ($\alpha = 2$) and maximally three-dimensional ($\alpha = 3$) tube families. The pairwise-angle multiset $\mathcal{A}(\mathcal{Q})$ of Definition 3 is computed in $O(N^2)$ time; results in §4.3 use the full computation.

B Proof of Proposition 5

This appendix supplies the derivation of the zero-curvature oscillatory representation. The argument is a Fourier reduction of the extended Dupire equation to its leading symbol, followed by a curvature audit.

Setting and additional regularity. Under Assumption 2, $C(k, \tau, \phi)$ is $C^{2,\varepsilon}$ on a compact frequency window $\mathcal{W} \subset \mathbb{R}^3$ and admits a smooth Fourier representation with amplitude $\widehat{C}$ supported in $\{|\xi| \sim R\}$, $R = 1/\delta$. We add a mild slowly-varying hypothesis on the local variance, standard in the local-volatility literature (cf. Dupire, 1994; Gatheral and Jacquier, 2014):

Assumption 12 (Slowly varying local variance on $\mathcal{W}$)**.** There exist $0 < \sigma_-^2 \leq \sigma_+^2 < \infty$ with $\sigma^2 \in [\sigma_-^2, \sigma_+^2]$ on $\mathcal{W}$, and $\|\nabla \sigma^2\|_{L^\infty(\mathcal{W})} = O(R^{-1/2})$.

Symbol reduction. The perpetual-augmented Dupire equation reads $\partial_\tau C = \frac{1}{2}\sigma^2 \partial_{kk} C - r_f(\phi)\partial_\phi C + \text{l.o.t.}$, with lower-order drift bounded by $O(R^{-1/2})$ under Assumption 12. Fourier-transforming and passing to characteristics yields the principal symbol $\xi_\tau = \frac{i}{2}\sigma^2(\text{ref})\xi_K^2 + r_f(\text{ref})\xi_\phi + O(R^{-1/2})$. Taking the real part governing the oscillatory phase, and absorbing σ^2 into the amplitude via the standard time-change, gives the reduced surface

$$\xi_\tau \;=\; \tfrac{1}{2}\xi_K^2 + r_f(\text{ref})\xi_\phi + O(R^{-1/2}), \tag{9}$$

which is the cone (3) of the main text with the $O(|\xi_K| \cdot |\xi_\phi|)$ correction absorbed into the linear $r_f(\text{ref})\xi_\phi$ under $|\xi_K| \sim R^{1/2}$.

Step 3: curvature audit and assembling (2). The Hessian of the defining function $F(\xi) = \xi_\tau - \frac{1}{2}\xi_K^2 - r_f(\mathrm{ref})\xi_\phi$ has eigenvalues $\{-1, 0, 0\}$; the single non-zero eigenvalue delivers the cone (as opposed to plane) geometry required by Theorem 1.2 of Bourgain et al. (2016), and the funding-axis correction is linear in ξ_ϕ and does not modify the Hessian — the cone curvature is inherited unchanged from the pure Dupire operator. Two caps at angular separation $\geq R^{-1/2}$ on the cone have tangent planes with angular separation $\geq cR^{-1/2}$, so the Bourgain–Demeter transversality condition holds. The $O(R^{-1/2})$ residual in (9) is absorbed into the $c(\varepsilon)$ constant via the standard perturbation lemma (Bourgain et al., 2016, Lemma 2.3). The amplitude a in (2) is the reduced $\widehat{C}$ times the standard change-of-variables Jacobian; the representation holds pointwise on the cone up to the $O(R^{-1/2})$ residual absorbed into the δ^ε loss of Theorem 8. $\square$

C Proof of Theorem 7

Wang's Kakeya volume inequality states that for any δ-tube family $\{T_i\}$ as in (1), in $\mathbb{R}^3$ with distinct directions, $|\bigcup_i T_i| \geq c_\varepsilon\, \delta^{3-\varepsilon} \sum_i |T_i|$ (Wang, 2025, Theorem 1.2). Let $\Phi : L^2(\mathcal{W}) \to L^2(\mathbb{R}^3)$ denote the forward oscillatory operator associated with the Dupire kernel via (2) restricted to the tube support $\mathcal{S}_\delta$. Cauchy–Schwarz against a directional test family $\{\psi_i\}$ supported on the tubes gives $\|\sigma - \hat{\sigma}_\delta\|_{L^2(\mathcal{W})} \geq |\langle \sigma - \hat{\sigma}_\delta, \Phi^*\psi\rangle| / \|\Phi^*\psi\|_{L^2(\mathbb{R}^3)}$. The numerator is bounded below by the volume $|\mathcal{S}_\delta| \gtrsim \delta^{3-\varepsilon}$; the denominator satisfies $\|\Phi^*\psi\|_{L^2(\mathbb{R}^3)} \lesssim \delta^{\alpha/2-\varepsilon}$ under $N_\delta(\mathcal{S}_\delta) \asymp \delta^{-\alpha}$, by Bourgain–Demeter transversality on the $\delta^{1/2}$ cap decomposition. Combining the two bounds and dividing delivers (4). The δ^ε loss is the same as in Theorem 8, both inherited from the induction-on-scales. $\square$

D Proof of Theorem 8

Application of the Bourgain–Demeter cone decoupling inequality (Bourgain et al., 2016) to the Dupire kernel of Proposition 5, combined with induction on scale to accumulate the intrinsic δ^{ε} loss. The effective constant decomposes as $c(\varepsilon) = C_{\mathrm{Dup}} \cdot C_{\mathrm{BDG}}(\varepsilon) \cdot C_{\mathrm{sc}}(\varepsilon) \cdot C_{\mathrm{ind}}(\varepsilon)$, where $C_{\mathrm{Dup}} \le \sigma_+^2/\sigma_-^2$ under Assumption 12; $C_{\mathrm{BDG}}(\varepsilon) = C_0\varepsilon^{-A}$ is the cone-decoupling constant of Bourgain et al. (2016); $C_{\mathrm{sc}}(\varepsilon) = C_0'\varepsilon^{-B}$ is the small-cap perturbation cost of Guth and Maldague (2022) for the $O(R^{-1/2})$ residual in (9); and $C_{\mathrm{ind}}(\varepsilon) = (1 + c_1\varepsilon)^{\lceil \log_2(1/\delta) \rceil} \le \delta^{-c_1\varepsilon/\log 2}$ is the scale-induction constant. The composed $\varepsilon^{-(A+B)}$ power is polynomial and absorbs into the δ^{ε} freedom. The proof consists of a routine dyadic caps decomposition of the cone (3) at width $\delta^{1/2}$, cap-wise Fourier inversion via (2), and stitching by Bourgain et al. (2016, Theorem 1.2). The absence of a $\log(1/\delta)$ dimensional factor — rather than the naive per-doubling log cost — is precisely why decoupling outperforms a straightforward telescoping argument: the induction is ε-bounded, not log-bounded. □

E Proof of Theorem 9

Under $\theta(\mathcal{Q}) \lesssim \delta^{1/2}$, the directional-tube family collapses into a slab of width $O(\delta^{1/2})$ in the (k, τ)-plane; the funding axis ϕ then carries no recoverable information at scale δ, and the recovery problem reduces to a two-dimensional Kakeya-tube problem in the (k, τ)-plane.

Reduction to the 2D Kakeya maximal function. Let $\Pi : \mathbb{R}^3 \to \mathbb{R}^2$ be the projection onto (k, τ). Under the sticky-regime hypothesis, for every quote q_i the tube $T_i(\delta)$ satisfies $T_i(\delta) \subset \Pi^{-1}(\widetilde{T}_i(\delta)) \cap \mathcal{W}$ for a 2D tube $\widetilde{T}_i(\delta)$ of length 1 and width $O(\delta)$, up to an $O(\delta^{1/2})$ slab error absorbed into the constant. The reconstruction L^2 error is then bounded below by the 2D Kakeya maximal-function bound: for any 2D tube family $\{\widetilde{T}_i\}$ of N tubes at scale δ pointing in directions separated by at least

$\delta^{1/2}$,

$$\Big\|\sum_i \mathbf{1}_{\widetilde{T}_i}\Big\|_{L^2} \gtrsim \delta \cdot N^{1/2}, \tag{10}$$

which is Wolff's improved 2D Kakeya maximal-function estimate (Wolff, 1995, Corollary 1).

From (10) **to the** $\delta^{1/2}$ **recovery bound.** Since $N_\delta(\mathcal{S}_\delta) \asymp \delta^{-\alpha}$ with $\alpha \in (2,3)$ in the sticky regime the exponent collapses to the 2D value $\alpha_{2\mathrm{D}} = 2$ up to a δ^{ε} loss. Substituting into (10) and inverting the covering-number identification of §2 gives $\|\sigma - \hat{\sigma}_\delta\|_{L^2(\mathcal{W})} \gtrsim \delta^{1/2}$, which is (6). The rate is attained by any regular one-dimensional smile estimator of $\sigma(k, \tau_0, \phi_0)$ in the log-strike direction at a fixed tenor and funding level, as is standard in the SVI / SSVI literature (Gatheral and Jacquier, 2014). The sticky-regime bound inherits its rate from Wolff's 2D Kakeya estimate (Wolff, 1995), not from Wang's 3D volume inequality (Wang, 2025), since the sticky regime is intrinsically two-dimensional; Wang's inequality enters only in Theorem 7's three-dimensional bound. $\square$

F Elasticity Derivation for the Log–Log Regression

The log–log regression (8) identifies an elasticity $\beta = \mathrm{d}\log \mathrm{RMSE}/\mathrm{d}\log\theta$, not a δ-exponent difference. Three admissible benchmarks apply. On the sticky branch, Theorem 9 with the sticky characterisation $\theta \lesssim \delta^{1/2}$ gives $\beta_{\text{sticky}} \approx +1$: a one-log increase in θ tracks a one-log increase in δ, so the sticky L^2 rate $\delta^{1/2}$ scales linearly in $\log\theta$. On the dispersed branch, Theorem 8's second-derivative rate $\delta^{1/p-1/2}$ for $p = 3$ with an identification lever $\theta \sim \delta^{\eta}$, $\eta \in (0, 1/2)$, gives $\beta_{\text{disp}}(\eta) = (1/p - 1/2)/\eta = -(1/6)/\eta$, negative in sign for every η but bounded in magnitude by $|\beta_{\text{disp}}| \leq 1/3$ over the admissible $\eta \geq 1/2$ neighbourhood of the crossover. The identification window of §4.5 — sign-flip-driven variation that approaches but does not fully cross the frontier — matches instead a *rate-difference* elasticity across the branches. Standard mixture arithmetic delivers an admissible negative-sign range $\beta_{\mathrm{rd}} \in [-1/2, -1/6]$: $-1/2$ corresponds to shocks that carry a book from deep sticky into the dispersed regime, and $-1/6$ to marginal, near-crossover shocks. The FE 95% CI $[-0.50, -0.02]$ overlaps this interval on its interior; the point estimate $\hat{\beta}_{\mathrm{FE}} = -0.10$ is consistent with the theory in sign and admissible in magnitude under this benchmark, though not a precise identification of the mixing weight.

G Data Manifest

Every reported figure and table is regenerated by a single top-level build invocation of the accompanying reproducibility bundle (`make all`). The audit manifest is a SHA-256 hash list of every source file and every produced artefact. Snapshot date: 2026-08-07. Bundle version: v5.0. Bundle contents: 46 raw files (13.07 MB) plus 13 analysis scripts, 7 auto-generated tables, 3 publication figures, and this manuscript. Every empirical claim carries a Level 1, Level 2, or Level 3 tag as set out in the bundle's data charter, and no Level 3 (grounded-simulation) item is used to support a headline result.

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